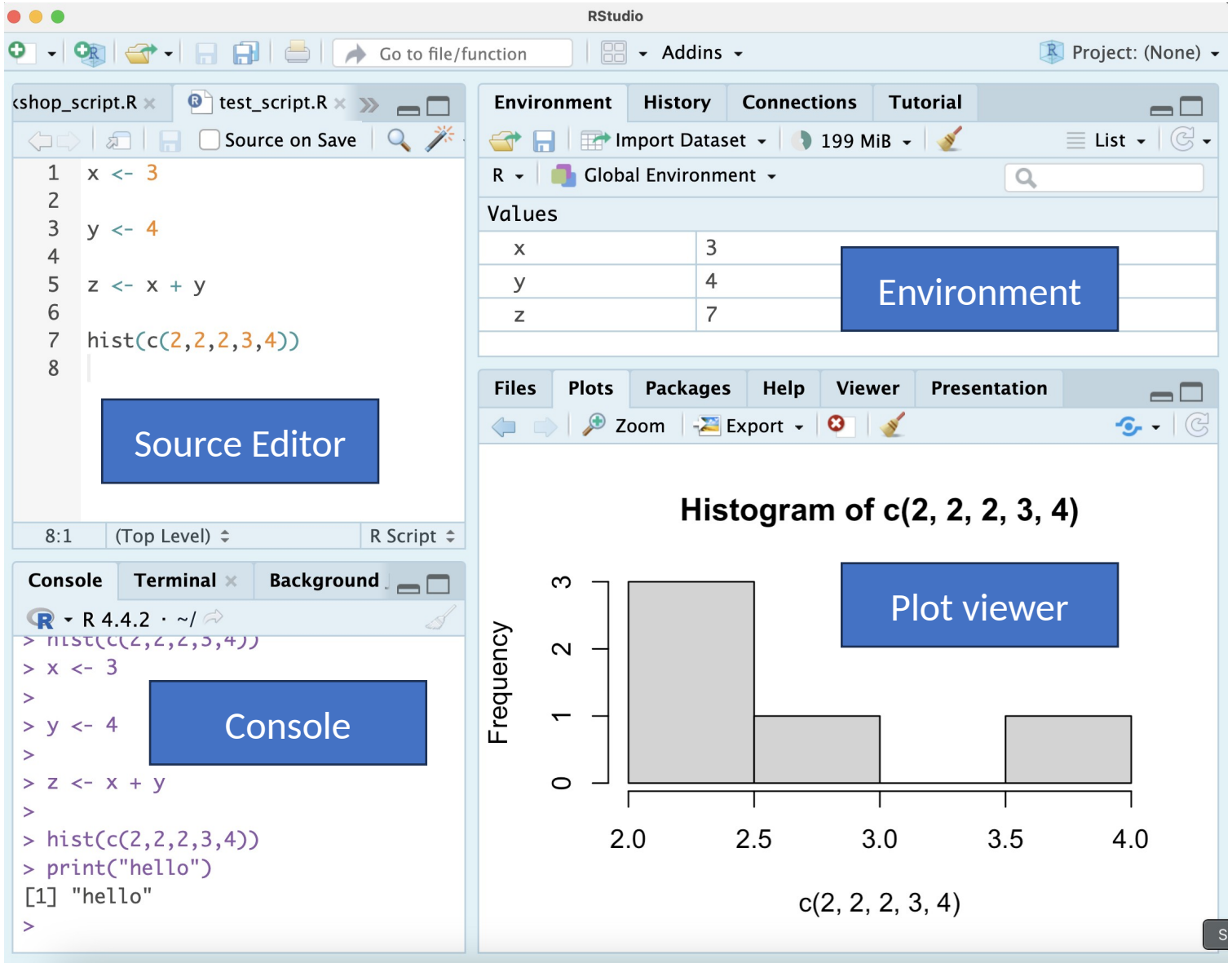
Core Knowledge Series: **Hands-On with Seurat**

Bioinformatics Core (BSR)

R Studio and R

- R is a powerful open-source programming language specifically designed for statistical computing, data analysis, and visualization. Many bioinformatics tools are developed in R.
- RStudio is user-friendly and makes coding in R easier with features like debugging, plotting, and environment.
- Packages native to R
  - Bioinformatics tools:
    - DESeq2 for RNA-Seq
    - Seurat for single-cell
    - Diffbind for ChIP and ATAC-seq
  - Data Manipulation
    - dplyr, tibble, tidyr, tidyverse

R Studio



- Source Editor (Top-Left)
  - Writing/editing R and Rmd scripts
- Environment (Top-Right)
  - Stores all created variables
- Console/Terminal (Bottom-Left)
  - Console: Executes R commands
  - Terminal: Navigate through files
- Files/Plots/Help (Bottom-right)
  - Mainly used for plot viewing. Can also navigate files and display help for functions.

Libraries in R

```
install.packages("Seurat")
library(Seurat)
```

- **Libraries** in R are collections of functions, datasets, and documentation that extend R's capabilities.
- Libraries must be **installed** and **loaded** before use

Basic syntax – Variables and Functions

```r
numbers <- c(1, 2, 3, 4, 5)
result <- sum(numbers)
print(result)
```

- **Object –** Fundamental data structure
  - Example: List/vector of numbers
- **Variable –** Name to store fundamental data structure
  - Example: “numbers” variable
- **Function** – Takes in input, performs operation(s), produces output
  - Sum function

Basic syntax – Variables and Functions

```r
numbers <- c(1, 2, 3, 4, 5)
result <- sum(numbers)
print(result)
```

- Variables are assigned using “<-” in R
- Functions often come before parenthesis with **arguments** inside
  - Inputs to a function that tell provide specifications for the function to perform its task
- sum(numbers)
  - Numbers is the argument to the sum function.
- How can we see what arguments a function takes?

Using the help operator

```
?sum
```

sum {base} R Documentation

Sum of Vector Elements

Description

`sum` returns the sum of all the values present in its arguments.

Usage

```
sum(..., na.rm = FALSE)
```

Arguments

`...`	numeric or complex or logical vectors.
`na.rm`	logical (`TRUE` or `FALSE`). Should missing values (including `NaN`) be removed?

Details

This is a generic function: methods can be defined for it directly or via the `Summary` group generic. For this to work properly, the arguments `...` should be unnamed, and dispatch is on the first argument.

If `na.rm` is `FALSE` an `NA` or `NaN` value in any of the arguments will cause a value of `NA` or `NaN` to be returned, otherwise `NA` and `NaN` values are ignored.

Logical true values are regarded as one, false values as zero. For historical reasons, `NULL` is accepted and treated as if it were `integer(0)`.

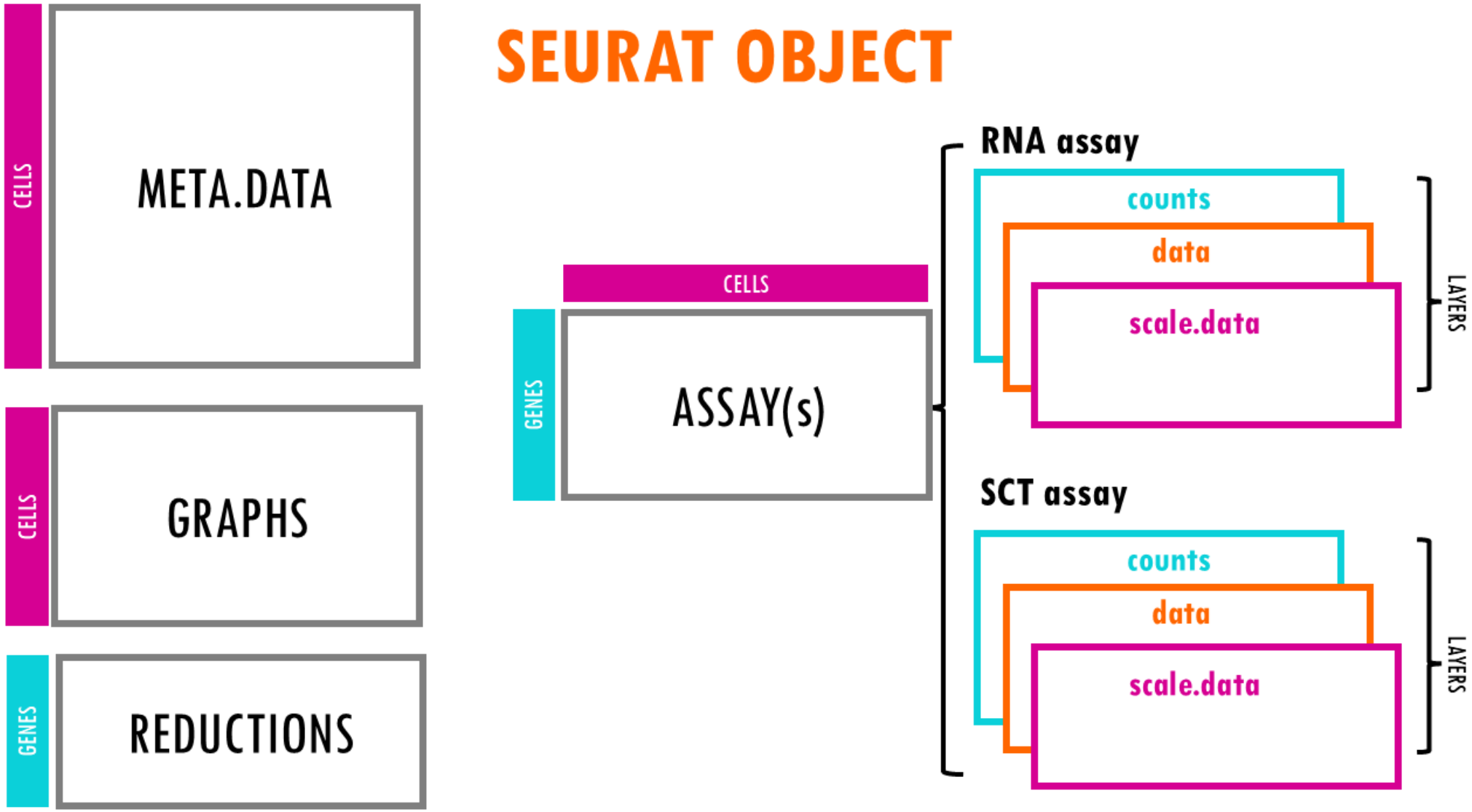
Loss of accuracy can occur when summing values of different signs: this can even occur for sufficiently long integer inputs if the partial sums would cause integer overflow. Where possible extended-precision accumulators are used, typically well supported with C99 and newer, but possibly platform-dependent.

- The help operator "?" can be placed in front of a function to open documentation for that function
  - Example: ?sum
- Note: The sum function has an optional parameter na.rm that will remove NA values if you specify TRUE
  - Default is na.rm = FALSE
- The ? operator is very helpful for Seurat functions
  - Well documented and often have many arguments

Seurat

- Seurat is an R package for analyzing single-cell RNA-seq data.
- It provides functions for normalization, clustering, plotting, differential analysis, integration, etc.

Seurat Object



Our PBMC Seurat Object

seurat_tutorial.Rmd | pbmc

Show Attributes

Name	Type	Value
pbmc	S4 [13714 x 2638] (SeuratObj	S4 object of class Seurat
assays	list [1]	List of length 1
meta.data	list [2638 x 6] (S3: data.frame	A data.frame with 2638 rows and 6 columns
active.assay	character [1]	'RNA'
active.ident	factor	Factor with 9 levels: "Naive CD4 T", "CD14+ Mono",
graphs	list [2]	List of length 2
neighbors	list [0]	List of length 0
reductions	list [2]	List of length 2
images	list [0]	List of length 0
project.name	character [1]	'pbmc3k'
misc	list [0]	List of length 0
version	list [1] (S3: package_version,	List of length 1
commands	list [7]	List of length 7
tools	list [0]	List of length 0

- Seurat objects contain slots. Think of them as compartments of the object.
  - Assays: Holds expression data
    - counts
    - data
    - scale.data
  - meta.data: Per-cell metadata
  - active.ident: Current identity of cells
    - Usually cluster number or cell type
  - reductions: Dimensionality reductions
    - PCA, tSNE, UMAP

R syntax with Seurat Object

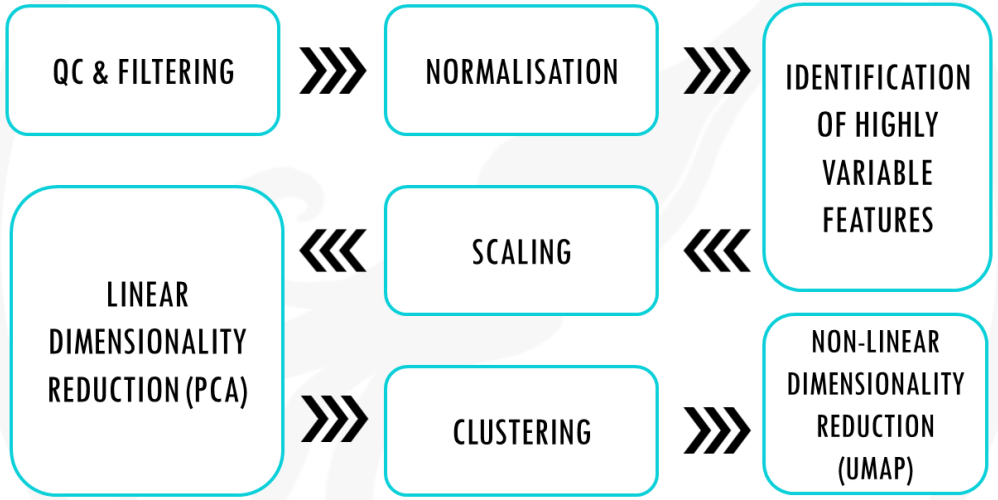
- $ operator is used to access metadata.
- @ operator is used to access other slots

meta.data	list [2638 x 6] (S3: data.frame	A data.frame with 2638 rows and 6 columns
orig.ident	factor	Factor with 1 level: "pbmc3k"
nCount_RNA	double [2638]	2419 4903 3147 2639 980 2163 ...
nFeature_RNA	integer [2638]	779 1352 1129 960 521 781 ...
percent.mt	double [2638]	3.02 3.79 0.89 1.74 1.22 1.66 ...
RNA_snn_res.0.5	factor	Factor with 9 levels: "0", "1", "2", "3", "4", "5", ...
seurat_clusters	factor	Factor with 9 levels: "0", "1", "2", "3", "4", "5", ...

```
> head(pbmc$percent.mt)
AAACATACAACCAC-1 AAACATTGAGCTAC-1 AAACATTGATCAGC-1 AAACCGTGCTTCCG-1 AAACCGTGTATGCG-1 AAACGCACTGGTAC-1
       3.0177759        3.7935958        0.8897363        1.7430845        1.2244898        1.6643551
```

```
> pbmc@assays
$RNA
Assay (v5) data with 13714 features for 2638 cells
Top 10 variable features:
 PPBP, LYZ, S100A9, IGLL5, GNLY, FTL, PF4, FTH1, GNG11, S100A8
Layers:
 counts, data, scale.data
```

Seurat workflow



- Create object – Read10X, CreateSeuratObject
  QC & filter – % mito, nFeature/nCount, subset
- Normalize – NormalizeData() or SCTransform()
- Variable features – FindVariableFeatures()
- Scale data – ScaleData() PCA – RunPCA(),
  inspect PCs (ElbowPlot)
- Clustering – FindNeighbors(), FindClusters()
  UMAP/tSNE – RunUMAP() / RunTSNE()
- Find markers – FindMarkers(),
  FindAllMarkers()
- Annotate – label clusters with known markers
- Save object – saveRDS()

Creating the Seurat Object

```r
library(Seurat)
library(dplyr)

# Creating folder and reading in data
outdir <- "~/single_cell_practice"
outdir <- path.expand(outdir)
if (!dir.exists(outdir)) {
  dir.create(outdir, recursive = TRUE)
}

data_file <- file.path(outdir, "pbmc3k_filtered_gene_bc_matrices.tar.gz")
download.file(
  url = "https://cf.10xgenomics.com/samples/cell/pbmc3k/pbmc3k_filtered_gene_bc_matrices.tar.gz",
  destfile = data_file,
  mode = "wb"
)
untar(data_file, exdir = outdir)
data_dir <- file.path(outdir, "filtered_gene_bc_matrices", "hg19")

################################################
pbmc.data <- Read10X(data.dir = data_dir)

pbmc <- CreateSeuratObject(
  counts = pbmc.data,
  project = "pbmc3k",
  min.cells = 3,
  min.features = 200
)
```

- The Seurat Object is the central container that stores
  - single-cell expression data
  - all downstream analysis results (QC metrics, clustering, embeddings, markers).
- Here, we read in the counts and create the object
- CreateSeuratObject is the function used.

Quality Control

```r
pbmc[["percent.mt"]] <- PercentageFeatureSet(pbmc, pattern = "^MT-")
head(pbmc[["percent.mt"]])
VlnPlot(pbmc, features = c("nFeature_RNA", "nCount_RNA", "percent.mt"), ncol = 3)
```

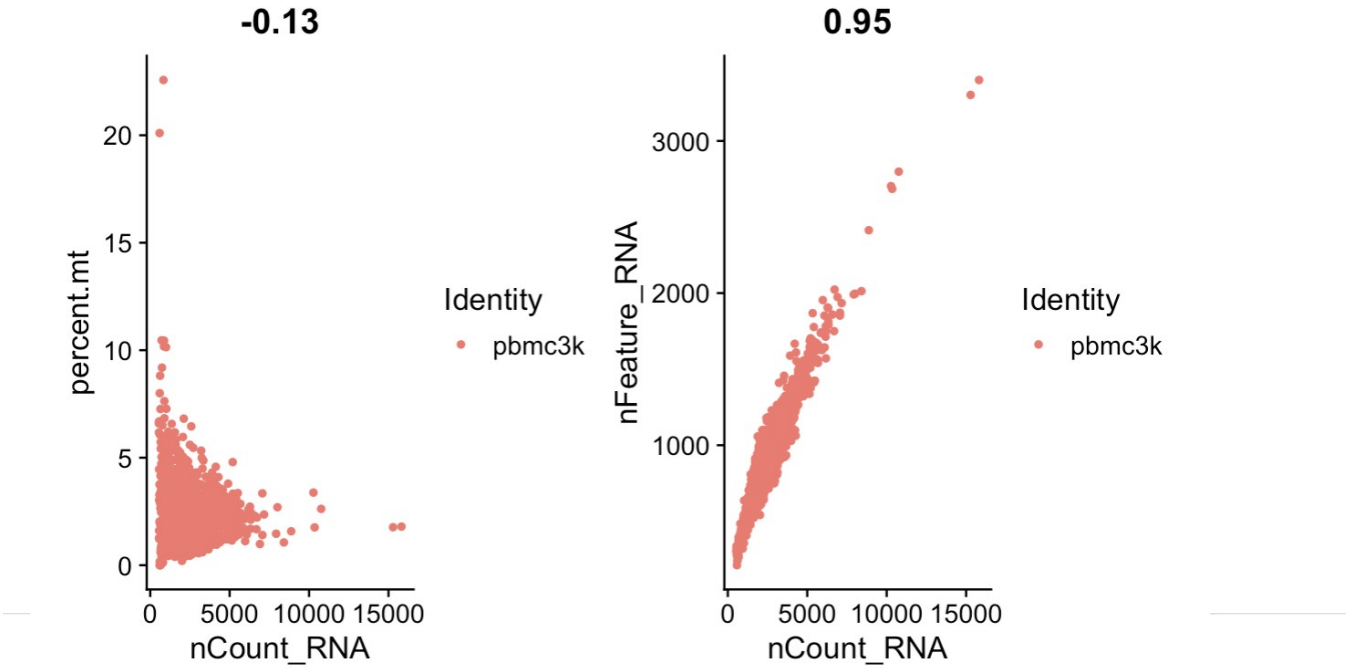
- Here, we add mitochondrial percentage as meta data
  - Note: Can use double brackets or $ to add to metadata
- The head() function gives the first few rows(cells)
- Seurat auto generates nFeature_RNA and nCount_RNA when creating the object
- High mitochondrial content could mean dying or low quality cells

Practice Question

- Use ?VlnPlot to get more information about the function
- Plot MALAT1 expression as a violin plot (no points shown).
  - Hint: Use the point size argument

Quality Control

```r
plot1 <- FeatureScatter(pbmc, feature1 = "nCount_RNA", feature2 = "percent.mt")
plot2 <- FeatureScatter(pbmc, feature1 = "nCount_RNA", feature2 = "nFeature_RNA")
plot1 + plot2
```



- Checking relationships between quality metrics can help detect
  - Low quality cells
  - Doublets
  - Outliers
- This helps us decide cutoffs for filtering

Filtering

```r
original_num_cells <- ncol(pbmc)
pbmc <- subset(pbmc, subset = nFeature_RNA > 200 & nFeature_RNA < 2500 & percent.mt < 5)
filtered_num_cells <- ncol(pbmc)
```

- The subset function is used to filter the cells
  - Number of genes expressed above 200
  - Number of genes expressed below 2500
  - Mitochondrial percentage < 5
- ncol() function will tell us how many cells are in the object
- Filtering is performed to remove low quality cells and artifacts

Practice Question

- Create variable pbmc2, where we filter pbmc using these parameters
  - Number of genes expressed above 500
  - Number of genes expressed below 2000
  - Percent mitochondria < 4
- How many more cells are in pbmc than pbmc2?
- What percentage of the cells were filtered out?

Solution

```r
# Filter to create pbmc2
pbmc2 <- subset(
  pbmc,
  subset = nFeature_RNA > 500 & nFeature_RNA < 2000 & percent.mt < 4
)

# Number of cells before and after
num_pbmc  <- ncol(pbmc)
num_pbmc2 <- ncol(pbmc2)

# How many more cells are in pbmc than pbmc2
num_diff <- num_pbmc - num_pbmc2

# Percent of cells filtered out
percent_filtered <- (num_diff / num_pbmc) * 100

# Print results
num_diff
# 318

percent_filtered
# 11.77778
```

Normalizing Data

```
```{r}
pbmc <- NormalizeData(pbmc, normalization.method = "LogNormalize", scale.factor = 10000)
```
```

- Counts need to be normalized so that expression can be compared across cells
  - Sequencing depth can differ between cells
  - Highly expressed genes can drown out biological signal
- Normalizing for sequencing depth:
  - Divide expression for all cells by their total counts and multiply by scale factor
- Reducing impact of highly expressed genes
  - Takes natural log of scale factored expression.
- This is all done with the NormalizeData() function

Finding Variable Genes

```r
pbmc <- FindVariableFeatures(pbmc, selection.method = "vst", nfeatures = 2000)

# Identify the 10 most highly variable genes
top10 <- head(VariableFeatures(pbmc), 10)

# plot variable features with and without labels
plot1 <- VariableFeaturePlot(pbmc)
plot2 <- LabelPoints(plot = plot1, points = top10, repel = TRUE)
plot1 + plot2
```

- Highly variable genes capture biological differences strongly
- Analyzing only variable genes reduces noise and helps with dimensionality reduction.

Scaling Data

```r
all.genes <- rownames(pbmc)
pbmc <- ScaleData(pbmc, features = all.genes)
```

- ScaleData() function scales the data so that each gene's expression is on the same scale.
  - Centers expression for each gene at 0
  - Divides by standard deviation of that gene's expression
- This is important for dimensionality reduction so that each gene has equal importance.

Dimensionality Reduction - PCA

```{r}
pbmc <- RunPCA(pbmc, features = VariableFeatures(object = pbmc))
```

- PCA reduces thousands of gene features into a few key components, making the data less noisy and easier to interpret.
  - Principal component: weighted combination of gene expression
  - PC1 explains the most variance in the data, PC2 explains the second most, etc.

Visualizations with PCA

```r
VizDimLoadings(pbmc, dims = 1:2, reduction = "pca")
```

```r
DimPlot(pbmc, reduction = "pca") + NoLegend()
```

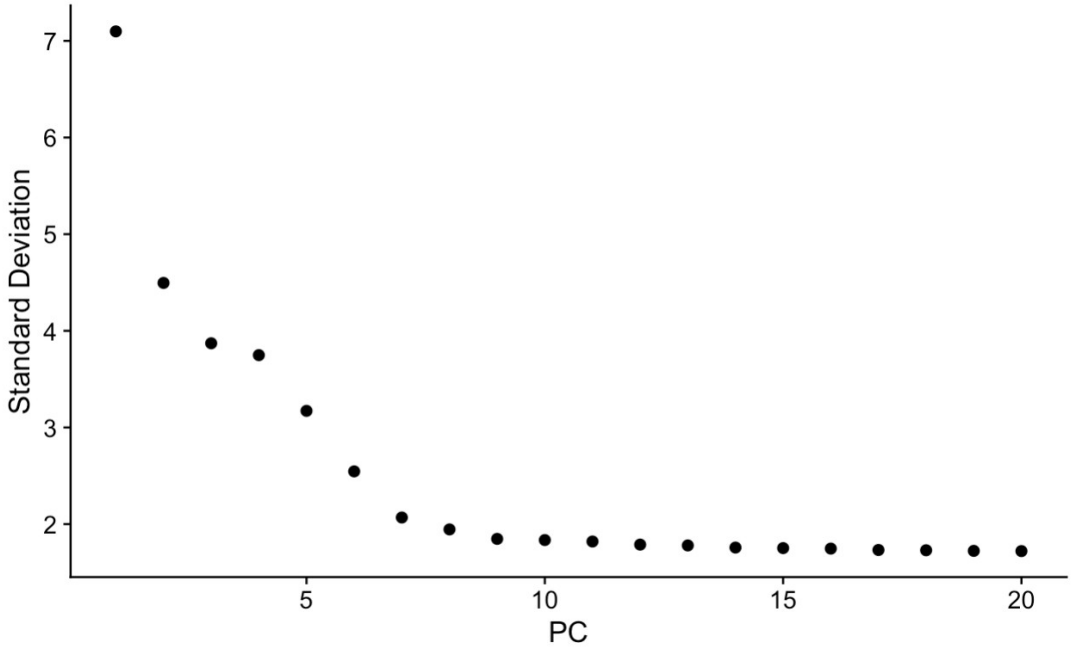
```r
DimHeatmap(pbmc, dims = 1, cells = 500, balanced = TRUE)
```

```r
ElbowPlot(pbmc)
```

- VizDimLoadings() – Plots the loadings/weights for the top genes for each principal component you specify
- DimPlot() will plot the cells on an axis using two principal components
- DimHeatmap() creates a heatmap using top genes from a principal component
- ElbowPlot() Plots the variance explained by each PC

Choosing how many PCs to use in downstream analysis

```
ElbowPlot(pbmc)
```



- The elbow plot can help you decide how many PCs to use.
- There looks to be an elbow at around 10 PCs.
- Majority of the true signal is likely captured in these first 10 PCs

Clustering the Cells

```{r}
pbmc <- FindNeighbors(pbmc, dims = 1:10)
pbmc <- FindClusters(pbmc, resolution = 0.5)
```

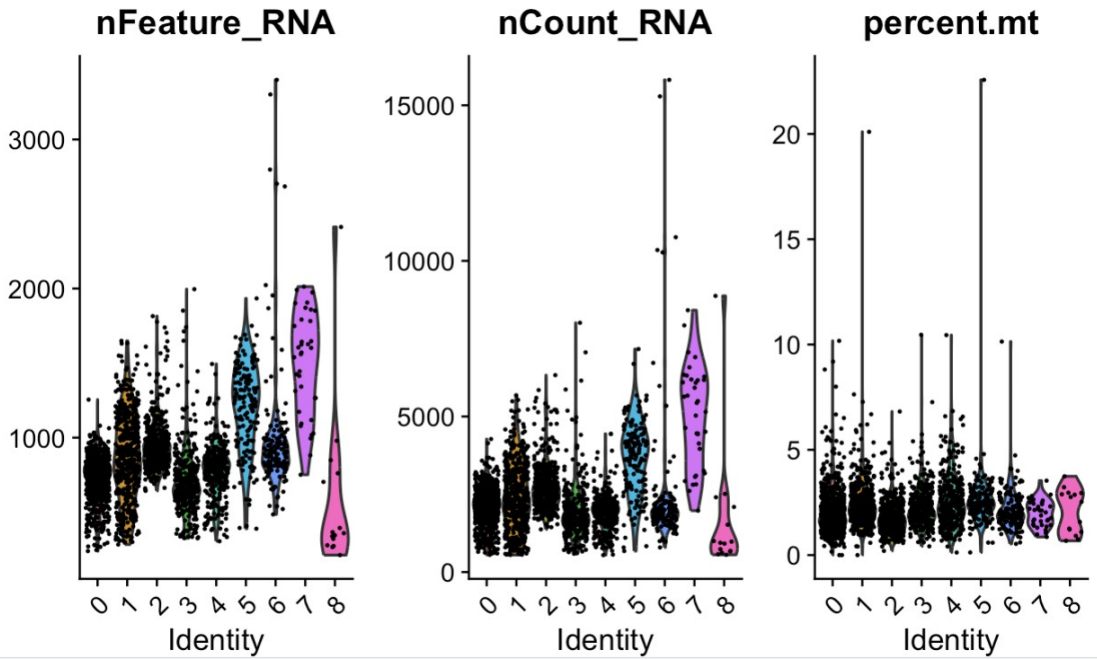
- FindNeighbors() builds a graph of cells based on how close they are in PCA space (who is nearest to who).
- FindClusters() then groups those cells into clusters the nearest neighbors graph.

Clustering the Cells

```r
pbmc <- FindNeighbors(pbmc, dims = 1:10)
pbmc <- FindClusters(pbmc, resolution = 0.5)
```

- Note that with FindNeighbors(), you need to specify which PCs to use.
  - Based on the elbow plot, we will use PCs 1-10
- With FindClusters, you need to specify resolution.
  - Higher resolution gives more clusters
  - Lower resolution gives less clusters
- Number of clusters depends on study goals/shape of UMAP/known markers that should cluster separately

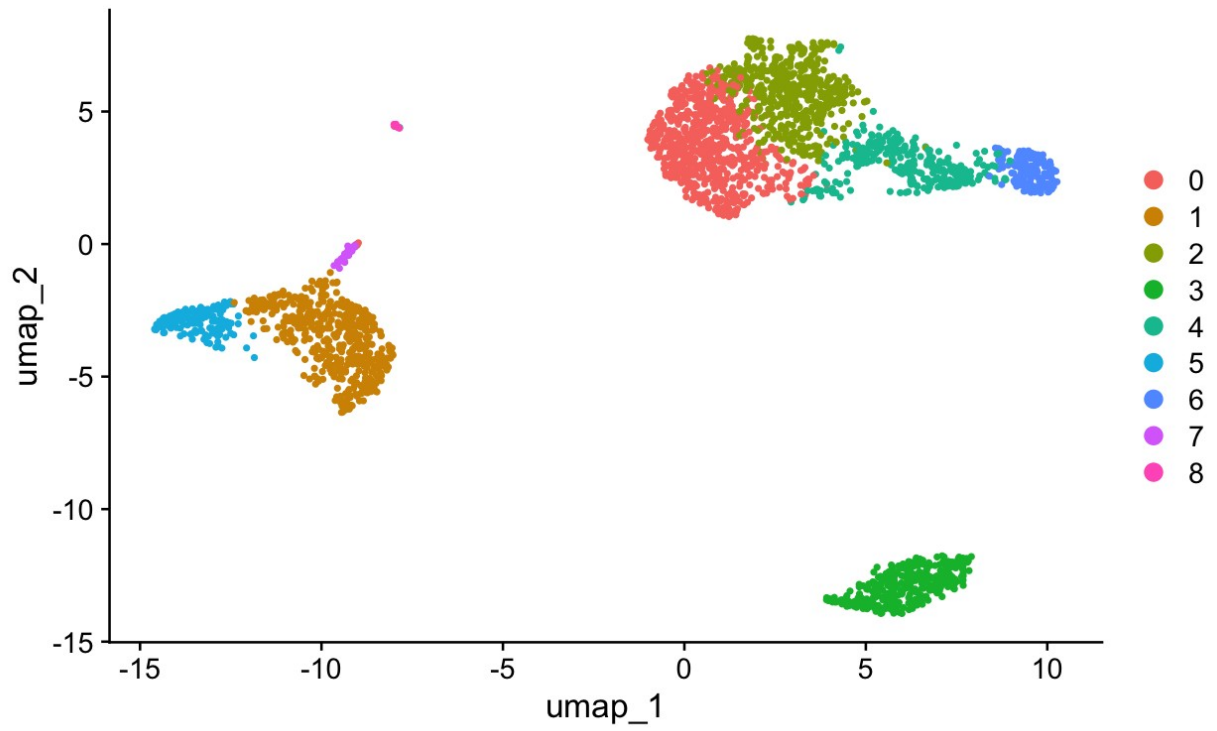
Clustering the cells



- Clustering the cells will change the active.ident (identity) of the cells
- The violin plot we made before is now split between cluster identities
- head(pbmc@active.ident) can tell us what the active identity is

UMAP – Non-linear dimensionality reduction

```r
pbmc <- RunUMAP(pbmc, dims = 1:10)
# note that you can set `label = TRUE` or use the LabelClusters function to help label
# individual clusters
DimPlot(pbmc, reduction = "umap")
```



- UMAP learns the structure of the data and projects it into 2D for visualization.
- It combines PCA-reduced information with neighbor relationships to preserve similarity between cells.

RunUMAP – min.dist and n.neighbors parameters

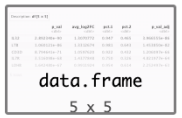
Usage

```
RunUMAP(object, ...)

## Default S3 method:
RunUMAP(
  object,
  reduction.key = "UMAP_",
  assay = NULL,
  reduction.model = NULL,
  return.model = FALSE,
  umap.method = "uwot",
  n.neighbors = 30L,
  n.components = 2L,
  metric = "cosine",
  n.epochs = NULL,
  learning.rate = 1,
  min.dist = 0.3,
```

- n.neighbors: neighbors used for graph.
  - Small = local detail
  - Large = global structure.
- min.dist: spacing of points.
  - Small = tight clusters
  - Large = spread out clusters.

Find Markers

```{r}
cluster2.markers <- FindMarkers(pbmc, ident.1 = 2)
head(cluster2.markers, n = 5)
```

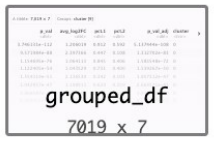


Description: df [5 × 5]

	p_val <dbl>	avg_log2FC <dbl>	pct.1 <dbl>	pct.2 <dbl>	p_val_adj <dbl>
IL32	2.892340e-90	1.3070772	0.947	0.465	3.966555e-86
LTB	1.060121e-86	1.3312674	0.981	0.643	1.453850e-82
CD3D	8.794641e-71	1.0597620	0.922	0.432	1.206097e-66
IL7R	3.516098e-68	1.4377848	0.750	0.326	4.821977e-64
LDHB	1.642480e-67	0.9911924	0.954	0.614	2.252497e-63

5 rows

```{r}
pbmc.markers <- FindAllMarkers(pbmc, only.pos = TRUE)

pbmc.markers %>%
    group_by(cluster) %>%
    dplyr::filter(avg_log2FC > 1)
```



A tibble: 7,019 × 7 Groups: cluster [9]

p_val <dbl>	avg_log2FC <dbl>	pct.1 <dbl>	pct.2 <dbl>	p_val_adj <dbl>	cluster <fctr>	gene <chr>
3.746131e-112	1.206019	0.912	0.592	5.137444e-108	0	LDHB
9.571984e-88	2.397366	0.447	0.108	1.312702e-83	0	CCR7
1.154695e-76	1.064113	0.845	0.406	1.583548e-72	0	CD3D

- FindMarkers tells you which genes are differentially expressed in a certain cluster
- FindAllMarkers tells you which genes are differentially expressed in all clusters

Feature Plotting

```{r}
# you can plot raw counts as well
VlnPlot(pbmc, features = c("NKG7", "PF4"), slot = "counts", log = TRUE)
```

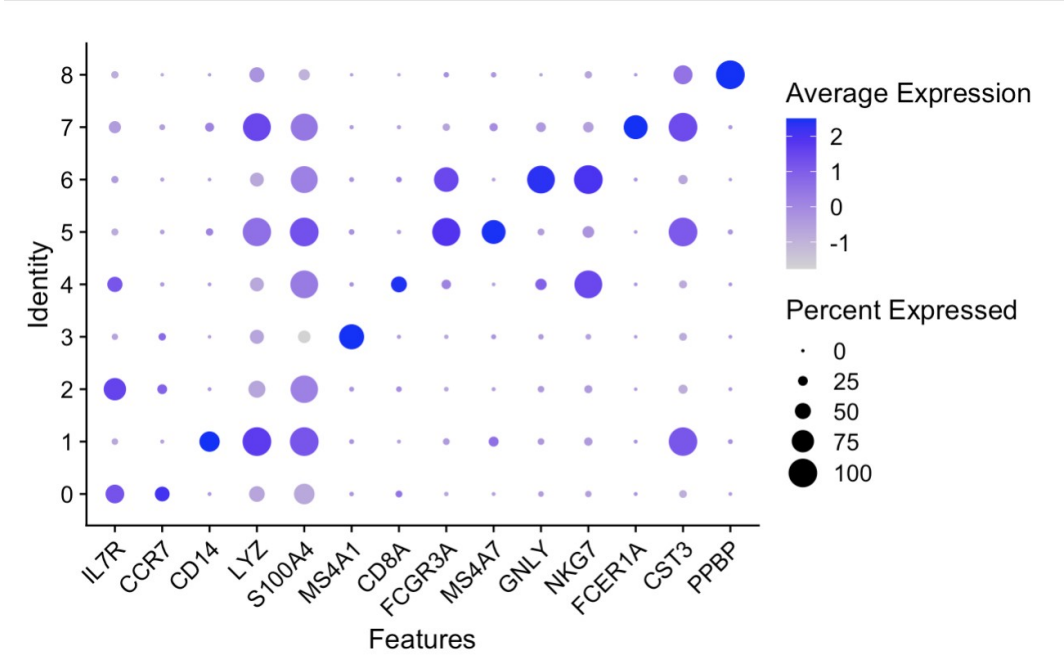
```{r}
FeaturePlot(pbmc, features = c("MS4A1", "GNLY", "CD3E", "CD14", "FCER1A", "FCGR3A", "LYZ", "PPBP",
    "CD8A"))
```

```{r}
pbmc.markers %>%
    group_by(cluster) %>%
    dplyr::filter(avg_log2FC > 1) %>%
    slice_head(n = 10) %>%
    ungroup() -> top10

DoHeatmap(pbmc, features = top10$gene) + NoLegend()
```

- You can plot certain features on violin plots, UMAP, and heatmap.
  - Specify features = c("Gene1", "Gene2",...)

Assigning Cell Types

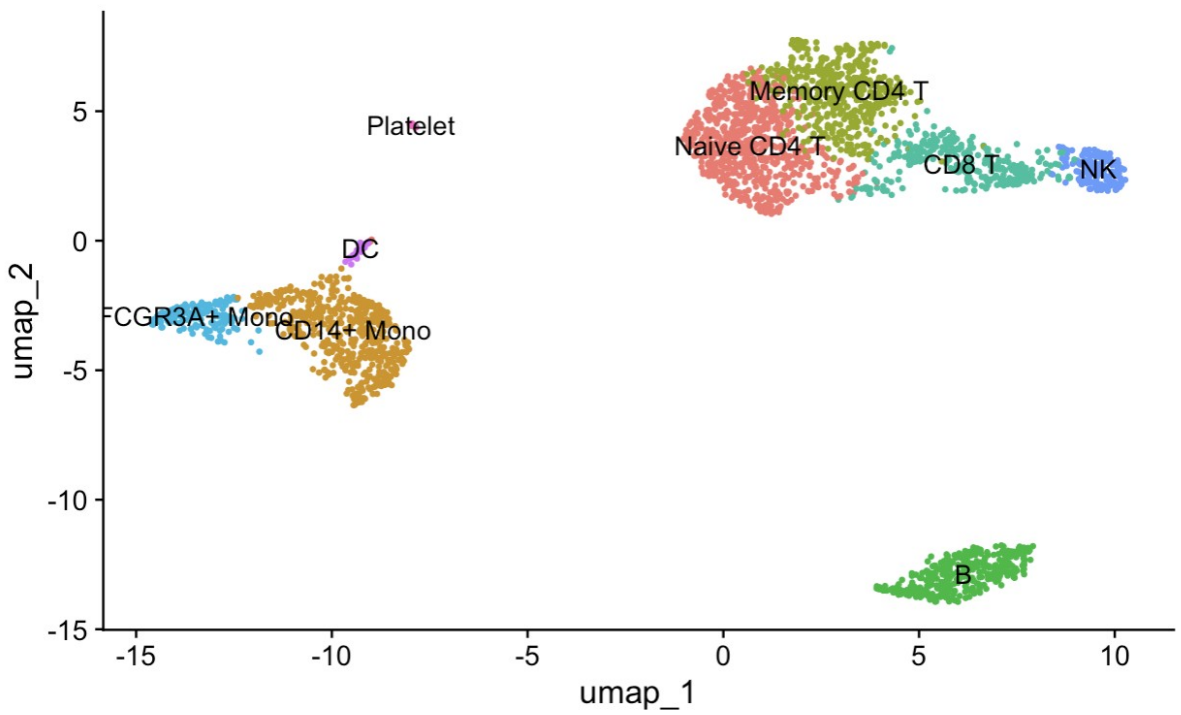
```r
# Define canonical markers by cluster
markers.to.plot <- c("IL7R", "CCR7",      # Naive CD4+ T
                     "CD14", "LYZ",       # CD14+ Mono
                     "S100A4",            # Memory CD4+
                     "MS4A1",             # B
                     "CD8A",              # CD8+ T
                     "FCGR3A", "MS4A7",   # FCGR3A+ Mono
                     "GNLY", "NKG7",      # NK
                     "FCER1A", "CST3",    # DC
                     "PPBP")              # Platelet

# Dot plot of marker expression across clusters
DotPlot(pbmc, features = markers.to.plot) +
  RotatedAxis()
```



- Dot plots are a good way to visualize expression of pre-defined cell-type markers

Assigning Cell Types

```r
new.cluster.ids <- c("Naive CD4 T", "CD14+ Mono", "Memory CD4 T", "B", "CD8 T", "FCGR3A+ Mono",
    "NK", "DC", "Platelet")
names(new.cluster.ids) <- levels(pbmc)
names(new.cluster.ids)
pbmc <- RenameIdents(pbmc, new.cluster.ids)
DimPlot(pbmc, reduction = "umap", label = TRUE, pt.size = 0.5) + NoLegend()
```



- The new cluster ids are stored as a list corresponding to cluster number.
- The names of the identity of the cells is then changed to reflect the cell type rather than the cluster number

Saving/reading object

```
```{r}
saveRDS(pbmc, file = "~/single_cell_practice/pbmc3k_final_annotated.rds")

obj <- readRDS("~/single_cell_practice/pbmc3k_final_annotated.rds")
```
```

- R objects can be saved as ".rds" files.
- saveRDS() takes in an object and a path
- readRDS() will read in an object from a path

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